

NCDC ASSESSMENT OF BIOLOGY

Focus of End of Cycle Assessment in Biology

Evaluate the learner's understanding and application of biological principles governing life processes in cells, plants, animals, and ecosystems, and the ability to analyse biological structures, functions, and interactions in order to make informed decisions that promote health, sustainable agriculture, biodiversity and environmental conservation.

Assessment Objectives

Assessment of Biology will be guided by four assessment objectives drawn from the four assessment constructs.

AO1: Evaluate the significance of the interactions that sustain life and energy production by analysing the structure of cells and tissues, the roles of biomolecules (water, lipids, proteins, and nucleic acids), and the processes of ATP production, cell division, and protein synthesis, to apply these biological principles to challenges in genetic technologies and health improvement.

AO2: Evaluate plant structure and physiology by analysing structural adaptations and photosynthetic pathways in C_3 and C_4 , environmental influences on photosynthesis, plant adaptations (to water availability), growth, photoperiodism, and the hormonal control of growth, to promote sustainable agricultural practices that improve crop yield and food security.

AO3: Evaluate the structure and physiology of animal sensory organs and systems (circulatory, nervous, immune, and homeostatic) by analysing blood circulation, gas transport, immunity (vaccination and allergies), neural transmission, sensory perception, homeostatic control, and the role of adaptive behaviours in survival, to make informed health decisions and promote animal welfare.

AO4: Evaluate the principles of genetics, evolutionary mechanisms, and ecological interactions by analysing Mendelian and non-Mendelian genetics, species evolution, speciation, resistance, extinction, population dynamics, ecosystem balance, and carbon emissions, to create sustainable strategies for managing invasive species, enhancing food security, and mitigating climate change.

CONSTRUCTS

Construct Name	Construct Description
1. Cellular Organisation, Respiration, and Molecular Analysis	Investigating the structure of cells, tissues, and levels of organisation, the roles of biomolecules (water, lipids, proteins, and nucleic acids), and processes of ATP production, cell division, protein synthesis, and genetic manipulation, by analysing how they interact to sustain life, and then ethically applying biological principles in genetic technologies and health improvement.
2. Plant physiology and adaptation	Evaluating plant structure and physiology by analysing structural adaptations and photosynthetic pathway differences in C ₃ and C ₄ , plant adaptations (to water availability), environmental influence on photosynthesis, growth, photoperiodism, and the hormonal control of growth, to promote sustainable agricultural practices that improve crop yield and food security.
3. Analysis of animal systems and behaviours in adapting to environmental changes for health, survival, and welfare	Evaluating the structure and physiology of animal sensory organs and systems (<i>circulatory, nervous, immune, and homeostatic</i>) by analysing blood circulation, gas transport, immunity (<i>vaccination and allergies</i>), neural transmission, sensory perception, homeostatic control, and how adaptive behaviours support survival to make informed health decisions and promote animal welfare.
4. Genetic, evolutionary, and ecological dynamics	Evaluating inheritance patterns, evolutionary mechanisms, and ecological interactions by analysing Mendelian and non-Mendelian genetics, species evolution, speciation, resistance, extinction, population dynamics, ecosystem balance, and carbon emissions, to inform sustainable strategies for managing invasive species, promoting food security, and mitigating climate change.

LINKAGE BETWEEN ASSESSMENT OBJECTIVES, CONSTRUCTS, AND SYLLABUS

Assessment Objective	Construct	Topics in the Syllabus
AO1	Cellular Organisation, Respiration, and Molecular Analysis	Topic 1. Cell Biology ; Topic 4. Respiration ; Topic 7. Inheritance and Evolution
AO2	Plant physiology and adaptation	Topic 2. Nutrition in plants ; Topic 6. Co-ordination ; Topic 8. Growth in Plants and Development in Insects
AO3	Analysis of animal systems and behaviours in adapting to environmental changes for health, survival, and welfare	Topic 3. Transport in Humans ; Topic 5. Homeostasis ; Topic 6. Coordination
AO4	Genetic, evolutionary, and ecological dynamics	Topic 7. Inheritance and Evolution Topic 8. Growth in Plants and Development in Insects Topic 9. Ecology

STRUCTURE OF THE ASSESSMENT PAPERS

There will be two assessment papers for Biology at the Advanced Secondary level.

Paper 1 (Theory) will contain two sections, A and B. The items in the paper will be scenario-based. The paper will take 3 hours.

Section A

This will have two compulsory items. One item will come from construct 1 (**Cellular organisation, respiration, and molecular Analysis**), addressing assessment objective 1, and the other from construct 2 (**Plant physiology and adaptation**), addressing assessment objective 2.

Section B

This will consist of parts I and II. The section will have four test items. Part I will have two items that come from construct 3 (**Organismal systems and homeostasis**), addressing assessment objective 3. Part II will have two items from construct 4 (**Genetic, evolutionary, and ecological dynamics**),

addressing assessment objective 4. A learner will respond to one item from each construct (Part).

Paper 2 (Practical) will contain two compulsory items and will take 3 hours. The items in the paper will be scenario-based.

- i)** Items in this paper will come from any of the four constructs addressing their respective assessment objectives
- ii)** One item will require the use of scientific investigations to solve a challenge presented in the scenario.
- iii)** The second item will require relating structural and behavioural mechanisms to the survival advantage of the organism.

Sample item from Paper 1 (theory)

Napak district is one of the agricultural areas in Uganda. It is a lowland area with high temperature, shallow soils that have limited organic matter and very low moisture. Farmers grow both native and non-native crops, however, they often get low yields from the non-native crops unlike from the native crops.

The farmers invited Agricultural Extension workers to study the situation and propose strategies to improve the yields. They conducted a study in the district and the data collected on native and non-native crops grown is shown in the table below.

Condition	Native crops e.g. Sisal, Aloe vera, wild millet	Non-native crops grown e.g. Amaranthus, Avocado trees, Wild peas
The type of leaf anatomy	Kranz anatomy present	Kranz anatomy absent
Distribution of stomata	Many stomata on lower leaf epidermis	Many stomata on upper leaf epidermis
Size of leaf cuticle	Thick	Thin
Number of hairs on the leaf	Numerous	Very few

You are part of the team invited by the Agricultural Extension Officer to help the farmers improve their crop yields.

Task: Account for the difference in the yields between native and non-native crops grown in the district, and propose strategies to enhance productivity and resilience.

Sample item from Paper 2 (Practical)

The Kita community members carry out commercial beetroot farming. However, the plants are regularly attacked by worms that destroy the leaves. The farmers use pesticide A to kill the worms, but the plants end up drying. This results in poor yields and financial losses to the farmers, and consequently, they have asked the Ministry of Agriculture for help.

A research team sent by the Ministry of Agriculture discovered that the pesticide is a reducing sugar, polar, and moves across the beetroot cell membranes into the cytoplasm, where it causes cell death after a week. One of the farmers reported that whenever she mixes the pesticide with solution D and sprays, the plants do not dry. The agricultural team wants to recommend mass use of solution D together with the pesticide; however, they do not have scientific justification for its effectiveness.

You have been provided with fresh beetroot, pesticide A, solution D, and a cork borer.

Scoring grid overview

PAPER	BASIS OF EVALUATION	SCORE 4	SCORE 3	SCORE 2	SCORE 1
THEORY	Interpretation: Comprehension of the task in respect to the scenario/context; Brief overview of the subject/problem. Presentation Generates ideas that address the task. Makes analysis of the context; Evaluates the context Presents ideas coherently Judgement Use the acquired knowledge to form an informed opinion/ conclusion or judgement. Uses acquired knowledge to form conclusions				
PRACTICAL	Planning; Risks and Mitigations; Procedure; Data Presentation; Analysis and Recommendation				

SAMPLE SCORING GRID (PRACTICAL PAPER 2)

Basis of Evaluation	Score 4	Score 3	Score 2	Score 1
Planning	Formulates a clear, testable aim ; develops a hypothesis linking liquid D to reduced beetroot membrane permeability; identifies all variables (<i>independent, dependent, controlled</i>) and lists all required materials . <u>Planning reflects a deep understanding of the investigation.</u>	States a clear aim and hypothesis; identifies most variables and materials; planning shows good understanding with minor omissions.	States a clear aim; identifies key variables and some materials; planning shows a basic understanding of the investigation.	States a general aim; identifies one or two variables or materials; planning shows limited understanding.

SAMPLE SCORING GRID (THEORY PAPER 1)

Basis of Evaluation	Score 4	Score 3	Score 2	Score 1
Interpretation of task/context (a) identifies the anatomical and ecological features of native and non-native plants grown (b) identifies Napak's environmental challenges	Accurately connects the task to biological concepts (<i>like adaptation, photosynthesis type, and water conservation</i>), clearly describes Napak's environmental challenges (<i>high temperature, low moisture, shallow soils</i>), and notes the differences in yield between native and non-native crops.	Links the task to some relevant biological concepts (e.g., C_4 vs C_3 pathway), describes the general environmental conditions, and notes differences in yields between native and non-native crops.	Mentions the environmental conditions that enable native crops perform better, but gives no or limited connection to biological concepts.	States that native crops grow better than non-native crops, but gives wrong or no environmental or biological basis.